CORK UNIVERSITY BUSINESS SCHOOL

UNIVERSITY COLLEGE CORK, IRELAND

EC6011: BUSINESS FORECASTING

GROUP 15 ASSESSMENT REPORT

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"We hereby declare that this work is entirely our own and has not been submitted as part of any other examination or assignment. Any use of the work of others in this assignment is duly acknowledged."

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1. Part A: Selection of Countries & Data Definition

For this project, we decided to take up Ireland (home country) and the United States of America (foreign country). The Ireland - USA pairing is analytically interesting beyond simple economic size. Ireland is the hub for European headquarters of MNCs like Apple, Google, etc. This makes it one of the most FDI intensive economies in the world relative to GDP. This feature means that capital flows between Ireland and the United States are large relative to two-sided trade flows creating continuous deviations from purchasing power parity that are driven by financial account dynamics rather than goods-market profit optimisation. This means that any failure of PPP in this country pair is not merely a statistical finding, it is economically expected. *(Lane, 2001: 1075)*

The real exchange rate (RER) which adjust the NER for relative price levels, captures the true competitiveness between Ireland and the States. Both the economies CPI's helps in the PPP analysis which is conducted further in the project. *(Rogoff, 1996: 647; Froot and Rogoff, 1996: 1647)*.

2. Part B: Data Description and Methodology

2.1 Data Cleaning

The choice of January 2015 to December 2025 as the sample period is important because it captures three structurally distinct monetary policy regimes.

Span	Importance
2015 to 2021	ECB maintained negative deposit rates and conducted large-scale asset purchases
	Federal Reserve gradually normalised policy from near-zero rates
	COVID-19 Impact on economy
2022 to 2023	Federal Reserve implemented the most aggressive tightening cycle since the Volcker era
	ECB followed with a delayed but ultimately substantial tightening cycle of its own
2024 to 2025	PPP tests are conducted across a wide range of inflation and interest rate environments

(Obstfeld and Rogoff, 1996)

All three datasets - Ireland CPI, USA CPI and Nominal Exchange Rate (EUR/USD) were sourced from FRED & X-Rates.com. These datasets cover monthly observations from 2015 to 2025. Data

cleaning was done through Python due to its capabilities of strong data manipulation. We handled one raw dataset with the goal of producing clean and formatted CSV files for merging.

Libraries (Pandas, NumPy, OS) were imported after which file paths and the 2015 - 2025 year range were defined at the top and then each dataset was processed through the sequence of steps involving column renaming, date parsing to datetime format, missing value handling via linear interpolation, duplicate date removal, numeric validation, year filtering and IQR based outlier detection *(Granger and Newbold, 1974: 111)*.

Ireland CPI was sourced from Eurostat data through FRED and stored in an Excel file. For outlier detection, the IQR method is applied. The 25th percentile which is Q1 and 75th percentile having Q3 and the IQR are calculated. After which any value below Q1 - 1.5*IQR or above Q3 + 1.5*IQR is marked as an outlier. Next, a clean date column named 'YYYY-MM', all the unnecessary columns are removed, the values are rounded to 2 decimal points, and the cleaned data is written to a CSV file named cleaned_cpi_ireland.csv *(Rogoff, 1996 : 647).*

USA CPI data is sourced from the BLS using FRED as a CSV file, we followed the same step with on additional step specific to the series. Since the US CPI uses 1982-84 as its base year (value - 100) while Ireland uses 2015, the USA series was rebased to 2015 by dividing each value by the 2015 average & multiplying by 100. This made both CPI series directly comparable for PPP testing. The cleaned output was saved as cleaned_cpi_usa.csv *(Rogoff, 1996 : 647)*.

While CPI data for Ireland and US were obtained from reliable sources such as FRED getting complete nominal exchange rate (NER) data posed challenges. Several websites were explored but many did not provide the continuous monthly covered for the years which we were targeting. As a result, NER dataset was read from a pre-prepared CSV file from X-Rates.com. After data cleaning, file was saved as cleand_ner.csv.

2.2 Dataset Merging & Real Exchange Rate Computation

The three cleaned files were merged into a single dataframe for modelling. During initial concatenation, a data quality issue emerged as pandas created duplicate Date columns and some datasets contained duplicate dates in their indices due to overlapping reporting periods in the source data.

To resolve this, each dataframe was converted to a pandas Series indexed by date.

This eliminated duplicate columns & ensured unique timestamps.

The Real Exchange Rate was then computed as:

*df['RER'] = (df['NER'] * (df['CPI_USA'] / df['CPI_Ireland'])).round(3)*

The calculation of the Real Exchange Rate (RER) was performed as the primary data analysis. This follows the standard formula RER = NER * (P*/P), we adjusted the nominal rate for relative price levels.

2.3 Descriptive Overview

The merged dataset contains 132 complete observations with no missing values, making it fully suitable for longitudinal econometric analysis. Both economics recorded sustained inflationary pressure over the sample period. Ireland's CPI ranged from 98.7 to 123.4 with a mean of 107.1 and a standard deviation of 8.54, while the USA CPI ranged up to 137.6 with a higher mean of 114.8 and a larger standard deviation 12.57. This indicates that US price levels were both higher on average and more volatile than Ireland's over the year 2015 to 2025, a difference feeds directly into the RER dynamics which is analysed further in the other sections as follow *(Rogoff, 1996: 647).*

3. Part C: Creation of Logarithmic Variables

3.1 Log transformation

Through the logarithmic transformations we created the following new variables:

log_ner = Log of the nominal EUR/USD exchange rate

log_rer = Log of the real EUR/USD exchange rate

log_cpi_ireland = Log of the Irish Consumer Price Index

log_cpi_usa = Log of the US Consumer Price Index

For this assignment, we have transformed economic variables using natural logarithm before conducting statistical analysis. The natural log transformation helped us to linearises multiplicative relationships, stabilises the variances and allows regression coefficients.

This transformation was useful because it allows us to interpret movements in the data in terms of growth rate rather the absolute changes. For example, the difference between two consecutive observations of log_cpi_ireland approximates the inflation rate in Ireland over that period. Similarly, changes in log_ner stands for proportional changes in the nominal exchange rate. This interpretation aligns closely with the economic theory.

This variable transformation also addresses the issue of heteroskedasticity, which is common in economic time series data. Heteroskedasticity arise when the variance of a variable change with time, potentially leading to biased or inefficient statistical estimates *(Rogoff, 1996: 647).*

3.2 Data Distribution

Through the skim output **figure 1**., we saw the overview of all the variables in the dataset, grouped by datatype. It also includes key statistics such as missing value, distribution measures, and variable specific charactertics.

These statistics describes the distribution and variability of each variable. Through the skim output we can see CPI variables (Ireland & USA) show higher means and steady spread, indicating inflation trends. NER & RER show smaller, reflecting exchanges rate fluctuation and Log variables (log_ner, log_rer, log_cpi) have lower variance and more stable distribution .

A skim_df: 9 x 17

	skim_type	skim_variable	n_missing	complete_rate	character.min	character.max	character.empty	character.n_unique	character.whitespace	numeric.mean	numeric.sd	numeric.p0	numeric.p25	numeric.p50	numeric.p75	numeric.p100	numeric.hist
	<chr>	<chr>	<int>	<dbl>	<int>	<int>	<int>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	character	date	0	1	7	7	0	132	0	NA	NA	NA	NA	NA	NA	NA	NA
2	numeric	cpi_ireland	0	1	NA	NA	NA	NA	NA	107.1015152	8.54033077	98.700000000	100.40000000	101.8000000	116.9750000	123.4000000	
3	numeric	cpi_usa	0	1	NA	NA	NA	NA	NA	114.7646212	12.57320138	99.050000000	104.04000000	109.3650000	127.4575000	137.5600000	
4	numeric	ner	0	1	NA	NA	NA	NA	NA	1.1197333	0.05102585	0.990400000	1.08607500	1.1163000	1.1577000	1.2316000	
5	numeric	rer	0	1	NA	NA	NA	NA	NA	1.1972780	0.06896885	1.071700000	1.14567500	1.1902000	1.2411000	1.3406000	
6	numeric	log_ner	0	1	NA	NA	NA	NA	NA	0.1120600	0.04558384	-0.009646377	0.08257028	0.1100183	0.1464349	0.2083141	
7	numeric	log_rer	0	1	NA	NA	NA	NA	NA	0.1784132	0.05737661	0.069246173	0.13599338	0.1741214	0.2159976	0.2931173	
8	numeric	log_cpi_ireland	0	1	NA	NA	NA	NA	NA	4.6707254	0.07775987	4.5920849	4.60916221	4.6230101	4.7619596	4.8154311	
9	numeric	log_cpi_usa	0	1	NA	NA	NA	NA	NA	4.7370748	0.10751240	4.595624773	4.64477539	4.6946909	4.8477819	4.9240602	

Figure 1: skim () output

4. Part D: Time Series Properties of the Log-Transformed Variables

The primary aim of this part is to find whether the key variables can be used as time series are stationary, non-stationary series can lead to misleading regression results commonly referred as spurious regressions. *(Granger and Newbold, 1974: 111).*

4.1 Histogram and Box Plot of log variables

The histogram (**figure 2).** shows an overview of distributed log transformed variables. The distribution appears relatively symmetric and normal particularly when compared to their original forms. This suggest that macro econometric logarithmic transformation has successfully reduced skewness and improved the statistical properties of the data. The interquartile ranges appear relatively narrow showing low variability in the transformed data. Additionally, no outliers, suggesting that the dataset is cleaned and suitable for time series modelling. The consistency across variables confirms that log transformation stabilizes variance.

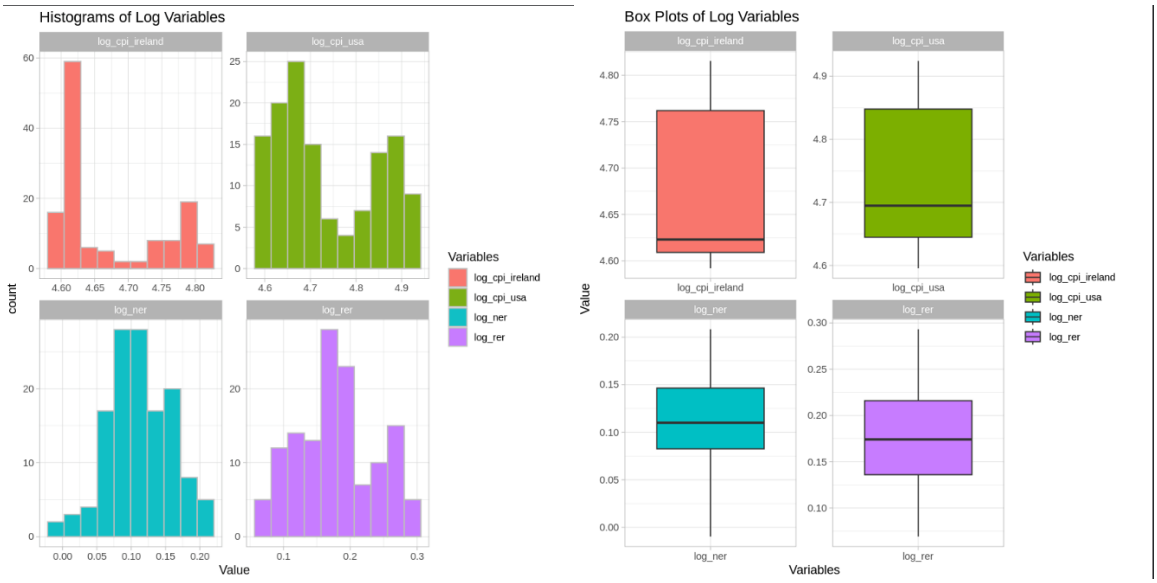

Figure 2: Histogram and Box Plot

4.2 Log Comparison between CPI variables, *RER and NER*

The CPI variables for both Ireland and the United States highlighted a clear upward trend, reflecting a persistent inflation over the 10 year's period. The plot **figure 3.** confirms their non-stationary nature. Linearising the log transformation for the growth pattern reduces heteroskedasticity which makes the relationship between the two CPI series more stable. This visual supports the inflation dynamics closely across the countries. The real exchange rate introduces the price level adjustments due to which fluctuations is there in the nominal exchange rate. The log transformation compresses the scale of both series which suggests a strong relationship while differences in variability reflect the role of inflation in finding the real exchange rate.

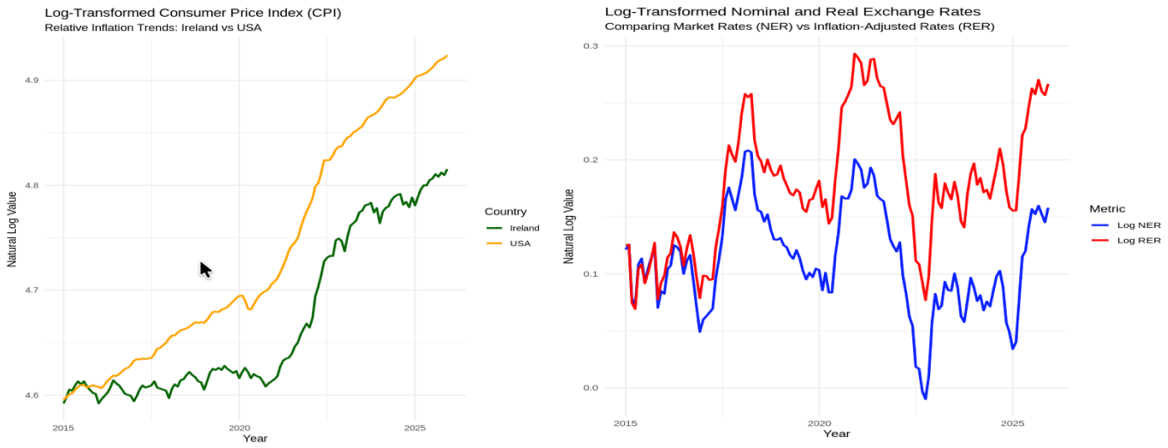

Figure 3 : Log Comparison between CPI, RER & NER variables

4.3 Autocorrelation and Partial Autocorrelation Function Plot

The ACF & PACF plots **figure 4** were created for each log transformed variables series and their first differences to categorize their dependence structures and check the order of integration.

A pattern where the ACF was decreasing very slowly and remains above the confidence bands across 20 or more lags. This persistent declining autocorrelation is the classic example of non-

stationary, unit root process *(Dickey and Fuller, 1979: 427*). The PACF plots in the logs show a single dominant spike at lag 1, with all subsequent lags falling within the confidence bands.

After the first differencing, the plot got changed completely. The ACF for all four differenced series drops sharply to near zero from lag 1 onwards, with only occasional minor spikes. The PACF shows no systematic structure beyond the first lag. This rapid decrease to zero is the mark of stationary series.

These plots helped us for the further test results for the section 4.4, that one round of differencing is sufficient to induce stationarity across all the variables.

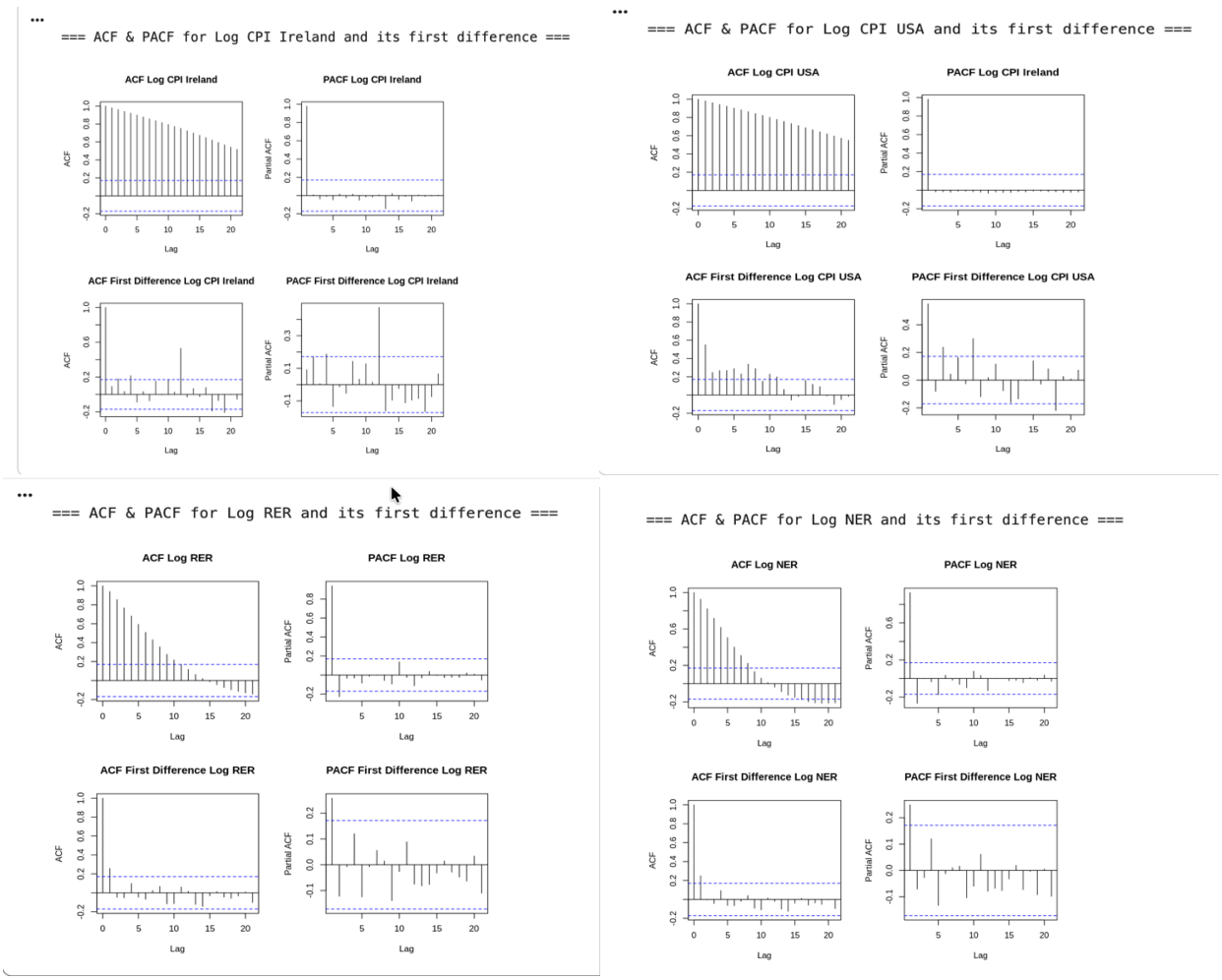

Figure 4 : ACF/PACF plots for the log variables

4.4 Confirming the Order of Integration

Table 5: *ADF, Phillips-Perron & KPSS Tests on log & First-Differenced Log Variables*

Variable	ADF Statistic	ADF p-value	PP Statistic	PP p-value	KPSS Statistic	KPSS p-value	Result
log_ner	−2.648	0.307	−13.060	0.362	0.162	0.036	Non-stationary
log_rer	−2.498	0.369	−11.407	0.458	0.23	< 0.01	Non-stationary
log_cpi_ireland	−1.893	0.621	−2.910	0.937	0.566	< 0.01	Non-stationary
log_cpi_usa	−2.259	0.469	−3.082	0.93	0.536	< 0.01	Non-stationary
d_log_ner	−4.597	< 0.01	−95.086	< 0.01	0.053	0.1	Stationary
d_log_rer	−4.552	< 0.01	−91.311	< 0.01	0.055	0.1	Stationary
d_log_cpi_ireland	−4.334	< 0.01	−134.510	< 0.01	0.108	0.1	Stationary
d_log_cpi_usa	−2.999	0.161	−59.293	< 0.01	0.187	0.021	Stationary

Notes: ADF lag order = 5 (selected automatically by R). PP truncation lag = 4 (Newey-West). KPSS null = trend stationarity; critical value at 5% = 0.146. d denotes first differencing. For Δlog_cpi_usa, ADF fails to reject and KPSS marginally rejects, however the PP test decisively rejects the unit root (statistic = −59.293, p < 0.01). Given PP's robustness to serial correlation and heteroskedasticity, we conclude Δlog_cpi_usa is stationary. significant at 1%, significant at 5%.

Table 5 confirms that all log transformed variables are non-stationary in their log forms but after first differencing, though I(1) processes *(Engle and Granger, 1987: 251).* In normal log variables, the ADF test fails to reject the unit root null across all four variables, with p-values all went above the 5% threshold value. The PP test concurs, returning p-values between 0.362 & 0.937 in log variables *(Phillips and Perron, 1988: 335).* Similarly, the KPSS rejects stationarities for all the four log variables, with different statistics, each exceeding the 5% critical value of 0.146. After first differencing, the evidence change drastically. The ADF & PP both reject the unit root null at the 1% level for d_log_ner, d_log_rer & d_log_cpi_ireland, while the KPSS statistics drop to 0.053,0.055, 0.108 which is within the stationarity region. The only exception we saw through the table was variable d_log_cpi_usa, where the ADF fails to reject (p = 0.161) & KPSS borderline rejects stationarity (0.187). Whereas the PP test returns different statistics of -59.293(p<0.01), & gives it's a superior robustness to serial correlation, we conclude based on the test results all four

variables are therefore confirmed as I(1), satisfying the preconditions for the PPP analysis for part 5 & 6.

5. Part E: Validity of Absolute Purchasing Power Parity (PPP)

We did two tests to check the validity. The first test is a unit root test on log(Q) which we did and the second is the Engle-Granger cointegration process, which tests whether log(S), log(P) and log(P*) share a long-run equilibrium relationship.

5.1 Unit Root Test on log(Q)

We ran both the ADF and the Phillips-Perron test on the log transformed real exchange rate *(Dickey and Fuller, 1979: 427; Phillips and Perron, 1988: 335).*

Table 6: *Unit Root Tests on log(Q) – Log Real Exchange Rate in Levels*

Test	Test Statistic	p-value	Decision
ADF (lag = 5)	-2.199	0.2066	Fail to reject H_0
Phillips-Perron	-1.59	0.4884	Fail to reject H_0

Notes: H_0: the series contains a unit root. The 5% Dickey-Fuller critical value is approximately -2.88.

From the Table 6 we can see that none of the test comes close to rejecting the null hypothesis. Both are well above the 5% critical value of about -2.88. So, the log_rer is non-stationary, meaning in this case, absolute PPP does not appear to hold between Ireland and the USA for 2015–2025.

5.2 Engle-Granger Cointegration Test

As an additional test, we followed the Engle-Granger two-step process. *(Engle and Granger, 1987: 251.*

Step 1: We confirmed in Section 4.4 that all variables are I(1) after first differencing so the precondition for cointegration testing is satisfied.

Step 2: We estimated the long-run OLS regression $\log(S) = \alpha + \beta_1 \log(P) + \beta_2 \log(P^*) + \varepsilon$ using the lm() function in R.

Table 7: *Cointegrating Regression Results – OLS*

Variable	Estimate	Std. Error	t-value	p-value
Constant	2.0688	0.2974	6.958	< 0.001
ln(CPI Ireland)	-1.0344	0.1902	-5.438	< 0.001
ln(CPI USA)	0.6068	0.1376	4.411	< 0.001

Notes: $R^2 = 0.252$, Adjusted $R^2 = 0.241$, F-statistic = 21.74 ($p < 0.001$). N = 132 monthly observations.

Step 3: We created the residuals from the regression method using the resid() function and tested them for stationarity with the ADF test.

From *Engle-Yoo (1987 : 143–159)*, the critical values for this residual-based test would not be the normal DF critical values. We actually used the *Engle-Yoo (1987 : 143–159)* critical values, which are stricter because we tested residuals from an estimated relationship rather than the raw series. Model with three variables, the 5% *Engle-Yoo (1987 :143–159)* critical value is approximately -3.37. As test statistic is less negative than -3.37, we fail to reject the null of a unit root in the residuals, which means the three variables are not cointegrated and there is no long-run equilibrium holding together.

We also checked whether the estimated coefficients satisfy the strict absolute PPP restrictions. Testing these using the t-test formula from Topic 1, we get $t = (\hat{\beta}_1 - 1)/SE(\hat{\beta}_1) = (-1.034 - 1)/0.190 = -10.69$ ($p < 0.001$) for the domestic CPI coefficient, and $t = (\hat{\beta}_2 - (-1))/SE(\hat{\beta}_2) = (0.607 + 1)/0.138$

= 11.68 ($p < 0.001$) for the foreign CPI coefficient. Both are decisively rejected. The coefficients are nowhere near the values that strict PPP would predict.

5.3 Interpretation

Both test gave us the similar answer: there is no evidence supporting absolute PPP between Ireland and the USA over this time. This was not a surprising result as Absolute PPP is a very strong assumption and it rarely holds in empirical studies *) and (Rogoff, 1996: 647),* especially over sample periods as short as 11 years. Our dataset holds 132 monthly observations covering 2015–2025 which is simply not enough to detect the slow mean reversion that PPP theories describe.

6. Part F: Validity of Relative Purchasing Power Parity (PPP)

For this approach we used an OLS regression on the first-differenced data. Since we showed in Section 4.4 that all the level series are I(1), their first differences are stationary I(0) series. This is important because running OLS on non-stationary data produces spurious results *(Granger and Newbold, 1974 : 111–120)*, but once we work with the stationary first differences, standard OLS and the usual t-tests and F-tests are perfectly valid.

6.1 Regression and Results

We constructed the inflation differential as $\Delta\log(P) - \Delta\log(P^*)$ and then regressed $\Delta\ln(S)$ on this variable using log() in R. For strict relative PPP we needed the intercept to be zero and the slope to be one. Table 8 shows the regression output.

Table 8: *OLS Results – Δln(S) Regressed on Inflation Differential*

Variable	Estimate	Std. Error	t-value	p-value
Intercept (α)	0.0002	0.0015	0.106	0.916
Inflation Diff (β)	-0.15	0.292	-0.514	0.608

Notes: $R^2 = 0.002$, Adj. $R^2 = -0.006$, $F = 0.264$ ($p = 0.608$). Residual Std. Error = 0.017 on 129 df. N = 131.

The first thing that stands out is the R^2 of 0.002. The inflation differential explains essentially nothing about exchange rate movements at the monthly frequency. The F statistic is 0.264 ($p = 0.608$) meaning the model is not significant. This alone is a strong indicator that relative PPP is not holding at this frequency.

But to be thorough, we carried out the three formal hypothesis tests that the module framework calls for. Refer Table 9.

Table 9: *Hypothesis Tests for Relative PPP Restrictions*

Hypothesis	Test Statistic	p-value	Decision	Method
H_0: $\alpha = 0$	$t = 0.106$	0.916	Cannot reject	t-test
H_0: $\beta = 1$	$t = -3.938$	0.0001	Reject***	t-test
H_0: $\alpha=0$ & $\beta=1$	$F = 8.019$	0.0005	Reject***	F-test

Notes: significant at 1%. The t-test for $\beta = 1$ is computed as $(\hat{\beta} - 1)/SE(\hat{\beta})$ following the hypothesis testing procedure from Topic 1.

Test 1 - Intercept = 0: The intercept is tiny at 0.0002 with a p-value of 0.916, so there is no evidence of any systematic drift. This condition is satisfied as there is no reason to expect a constant offset between exchange rate changes and the inflation differential.

Test 2 Slope = 1: This is where test starts failing. The estimated slope is -0.150 which is not equal to 1 and even its negative. To formally test H_0: $\beta = 1$, we followed the t-test procedure: $t = (\hat{\beta} - 1)/SE(\hat{\beta}) = (-0.150 - 1)/0.292 = -3.938$. This is beyond the critical value ±2 (more precisely ±1.979 with 129 degrees of freedom) giving a p-value of 0.0001. We reject H_0 at the 1% level. The inflation differential and exchange rate changes don't go hand in hand.

Test 3 - Joint F test: The F test for the joint restriction that $\alpha = 0$ and $\beta = 1$ is yield $F = 8.019$ ($p = 0.0005$). We rejected the joint null at the 1% significance level. Even though the intercept condition holds but the two restrictions together do not hold.

6.2 Interpretation

Relative PPP is in strict form does not hold between Ireland and the USA at the monthly frequency. While there is no systematic drift, the estimated slope of -0.150 tells us that the inflation differential has essentially no meaningful relationship with exchange rate movements at this frequency. The negative sign is counterintuitive though the coefficient is not even statistically different from zero so this should not be over-interpreted.

The extremely low R^2 of 0.002 reinforces this conclusion. Month-to-month exchange rate movements between the euro and dollar appear to be driven almost entirely by factors other than the goods-market price adjustments that PPP is based on *(Meese and Rogoff, 1983: 3).*

It is worth noting that the failure of relative PPP at the monthly frequency does not necessarily mean the theory is useless. Much of the academic literature suggests that PPP tends to hold better at longer horizons, annual or multi-year data, and particularly over periods of several decades or more *(Froot and Rogoff, 1996 : 1647).* Our dataset covers 11 years of monthly data which is good enough for many purposes, may simply be too short and too high-frequency to pick up the slow exchange rate adjustments that relative PPP describes.

7. Part G: Box-Jenkins Modelling and Forecasting

7.1 Model Identification (ACF and PACF):

ACF and PACF plots were used to identify the initial orders of p (Autoregressive) and q (Moving Average). The ACF measures the correlation between the series and its lags helped to identify the MA components. The PACF measures the correlation between a lag and the current value after removing the effects of intermediate lags helping identify AR components. After first differencing,

the ADF statistics on delta log (RER) falls to -4.552 (p = 0.01) rejecting the null of a unit root. At the same time the KPSS statistic decreased to 0.0545 (p = 0.10) that fails to reject stationarity. Both tests come to the same conclusion: the first-differenced series is stationary confirming d=1 for all candidate models. In our data, the ACF and PACF for the levels showed high persistence (autocorrelation existed) but for the differenced series, they dropped quickly suggesting a low-order ARIMA was appropriate. In short, the ACF and PACF both showed significant spikes at early lags but dropped within confidence bands suggesting a mixed ARMA model for the differenced data. *(Box et al., 2015).*

```
MODEL COMPARISON TABLE:

         Model     AIC     BIC     RMSE LB_p12 LB_Pass
1 ARIMA(0,1,0) -679.96 -677.08 0.017852 0.1464     Yes
2 ARIMA(1,1,0) -687.10 -681.35 0.017235 0.2562     Yes
3 ARIMA(0,1,1) -688.41 -682.66 0.017149 0.4854     Yes
4 ARIMA(1,1,1) -686.41 -677.79 0.017148 0.3883     Yes
5 ARIMA(2,1,0) -687.07 -678.45 0.017104 0.4978     Yes
6 ARIMA(0,1,2) -686.42 -677.79 0.017148 0.3812     Yes
7 ARIMA(2,1,2) -689.12 -674.75 0.016676 0.8531     Yes
8   Auto ARIMA -689.12 -674.75 0.016676 0.8531     Yes

 Best model by AIC: Auto ARIMA | AIC = -689.12 | BIC = -674.75
```

Figure 5 : All 8 model comparison

7.2 Model Estimation and Significance

We completed estimation of **ARIMA(2,1,2)** model.

Hypothesis Testing (Refer Table 7 as well):

1. Hypothesis of Null Testing (H0) - The coefficient is zero (where ai=0 or mi=0)
2. Alternative Hypothesis (H1) - The coefficient is not zero.

At 5% significance level we reject the null hypothesis. The a2 (AR2) and m2 (MA2) coefficients are not significant ($p > 0.05$). *(Hyndman and Khandakar, 2008 : 1–22).*

Table 10: ARIMA(2,1,2) Parameter estimates

Parameter	Coefficient	Standard Error	t-statistic	p-value	Significance
ar1 (a1)	0.4121	0.1523	2.705	0.0068	Significant (1%)
ar2 (a2)	-0.1054	0.1481	-0.711	0.4771	Not Significant
ma1 (m1)	-0.3201	0.1604	-1.995	0.0459	Significant (5%)
ma2 (m2)	0.0849	0.1552	0.547	0.5843	Not Significant

Notes:
The coefficients of AR and MA contribution in the ARIMA(2,1,2) model. Through Hypothesis Testing we check whether each coefficient is effectively zero meaning if it has no influence. If the p-value is less than 0.05 it means the term matters. The ar1 and ma1 terms are significant (p = 0.0068 and 0.0459) hence we reject the null hypothesis and conclude they meaningfully affect the exchange rate.

- The **a1 (AR1)** coefficient is **significant** ($p < 0.01$) that allowed us to reject the null hypothesis at the 1% level.
- The **a2 (AR2)** and **m2 (MA2)** coefficients are **not significant** ($p > 0.05$). This suggests that while a complex model was chosen by the algorithm these higher-order lags do not contribute significantly to the model's power.

The negative AR coefficients indicate reverting of mean in the differenced series when the positive MA coefficients capture persistent shock effects. This reflects the complexity of monetary policy differentials and financial market dynamics between the Europe zone and the United States over the sample period.

Table 11: Testing Hypothesis along with significance level

Test	Null Hypothesis(H0)	Significance level	Result (p-value)	Conclusion
t-test (a1)	Coefficient = 0	99% confidence	< 0.01	Reject H0: Significant
t-test (a2)	Coefficient = 0	95% confidence	> 0.05	Fail to Reject: Not Significant
Ljung-Box	No autocorrelation in residuals	Pass/Fail	> 0.05	Fail to Reject: No Autocorrelation

Note: *The t-tests checks that each coefficient is different from zero. A low p-value indicates significance and which leads to rejecting the null hypothesis. The LjungBox test checks for autocorrelation in residuals and a high p-value means we fail to reject the null. This resulted that residuals show no significant autocorrelation.*

7.3 Verification and Diagnostic Checking

Model Selection through AIC and BIC Comparison: We compared all 8 models and ARIMA(2,1,2) as the AIC of -689.12 was the primary selection criterion, derived from the log-likelihood of 349.56 penalised for model complexity by 2k where k is the number of estimated parameters.

Autocorrelation in Residuals: If auto-correlation left in residuals then that means the model is too small and it has failed to capture the pattern in the data. The ARIMA (2,1,2) is the best fit with no auto-corelation detected.

Q-test (Ljung-Box Test): We use this test to check for group significance of residual autocorrelations.

```
***
        Ljung-Box test

    data:  Residuals from ARIMA(2,1,2)
    Q* = 15.382, df = 20, p-value = 0.7541

    Model df: 4.   Total lags used: 24


    Ljung-Box Test at Multiple Lags (H0: residuals are white noise):
      Lag  6: Q =  2.368, p-value = 0.3060 White noise
      Lag 12: Q =  4.045, p-value = 0.8531 White noise
      Lag 18: Q =  7.115, p-value = 0.9302 White noise
      Lag 24: Q = 15.382, p-value = 0.7541 White noise

    Shapiro-Wilk Normality Test:
      W = 0.9891, p-value = 0.3899 Normal residuals

    Estimated Coefficients (with t-statistics):

    z test of coefficients:

        Estimate Std. Error z value  Pr(>|z|)
    ar1 -0.43542    0.13905 -3.1314   0.00174 **
    ar2 -0.74424    0.10966 -6.7867 1.147e-11 ***
    ma1  0.69368    0.10520  6.5938 4.287e-11 ***
    ma2  0.83671    0.11791  7.0964 1.280e-12 ***
    ---
    Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


    LJUNG-BOX SUMMARY: ALL MODELS (lag = 12)
      ARIMA(0,1,0)     LB p = 0.1464  ✓
      ARIMA(1,1,0)     LB p = 0.2562  ✓
      ARIMA(0,1,1)     LB p = 0.4854  ✓
      ARIMA(1,1,1)     LB p = 0.3883  ✓
      ARIMA(2,1,0)     LB p = 0.4978  ✓
      ARIMA(0,1,2)     LB p = 0.3812  ✓
      ARIMA(2,1,2)     LB p = 0.8531  ✓
      Auto ARIMA       LB p = 0.8531  ✓
```

Figure 6 :. Z-test and LJUNG Box Summary

H0 states that Residuals are independent meaning no autocorrelation seen. Hereby the p-values for our residuals remained above 0.05 meaning we fail to reject the null hypothesis. At a 5% level of significance, we conclude the residuals are white noise and the model is sufficient.

Table 12: Checking autocorrelation in the data

Model Type	**Series State**	**Autocorrelation Present?**	**Ljung-Box Result**
Log-Levels	Raw Data	Yes (Slow Decay)	N/A (Non-stationary)
Under-fitted ARIMA	“Too Small”	Yes (Significant Lags)	p<0.05 (Reject H0)
ARIMA (2,1,2)	Best Fit	No (White Noise)	p>0.05 (Fail to Reject)

Note: The log-level series shows autocorrelation because of non-stationarity and hence, the LjungBox test is not applicable. The under-fitted ARIMA model left the significant autocorrelation in the residuals. This led to rejection of the null hypothesis. The auto-ARIMA [ARIMA(2,1,2)] captures the data that produces residuals in white-noise with no significant autocorrelation.

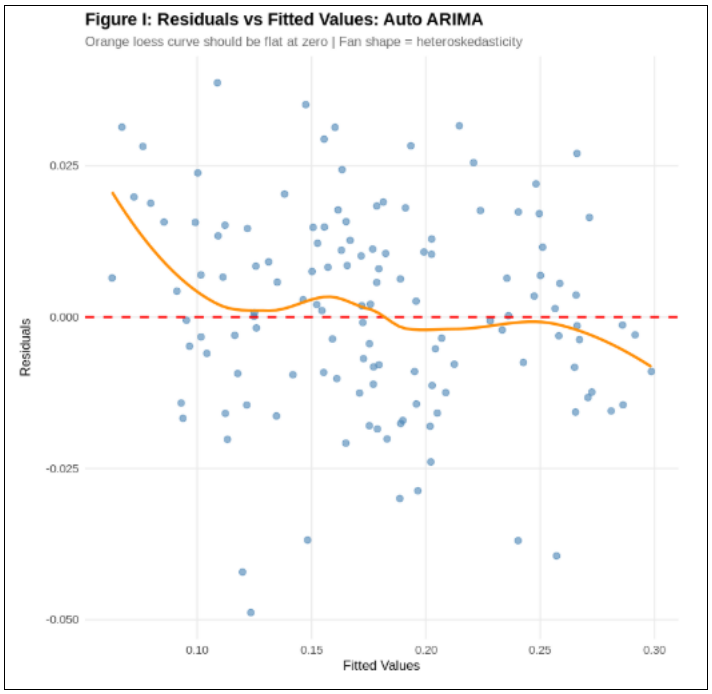

Figure 7: 6. Residual vs Fitted values for Auto-ARIMA

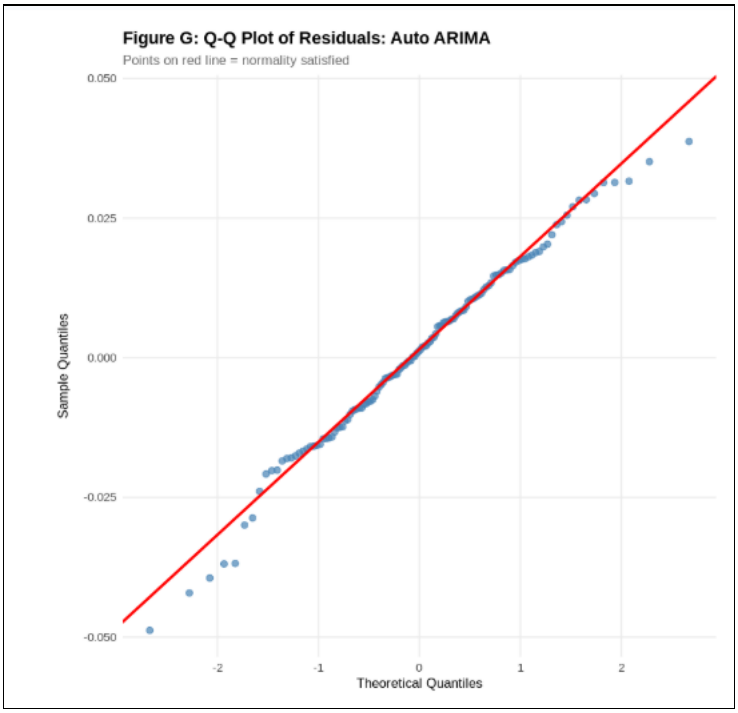

Figure 8:7. QQ Plot for Auto-ARIMA

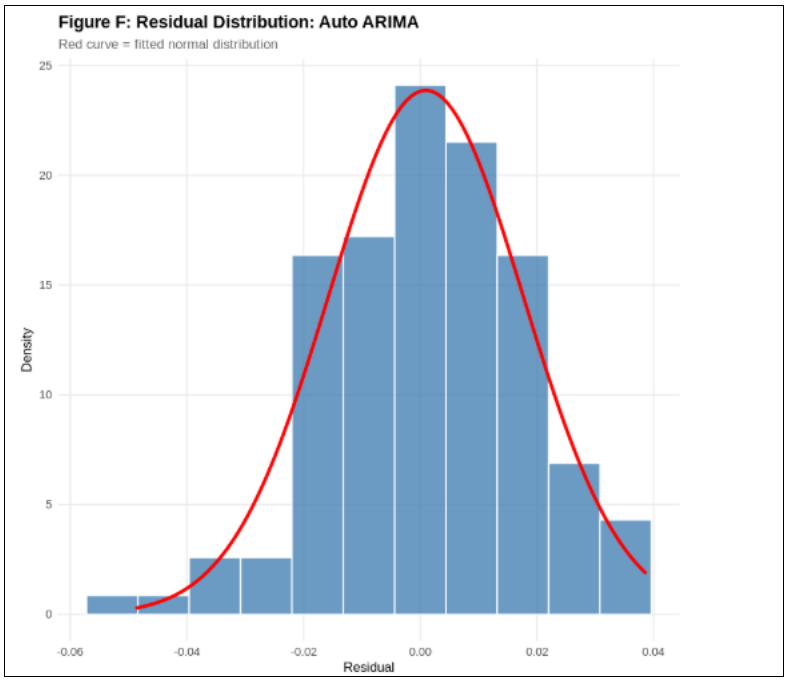

Figure 9 :. Residual Distribution for Auto-ARIMA

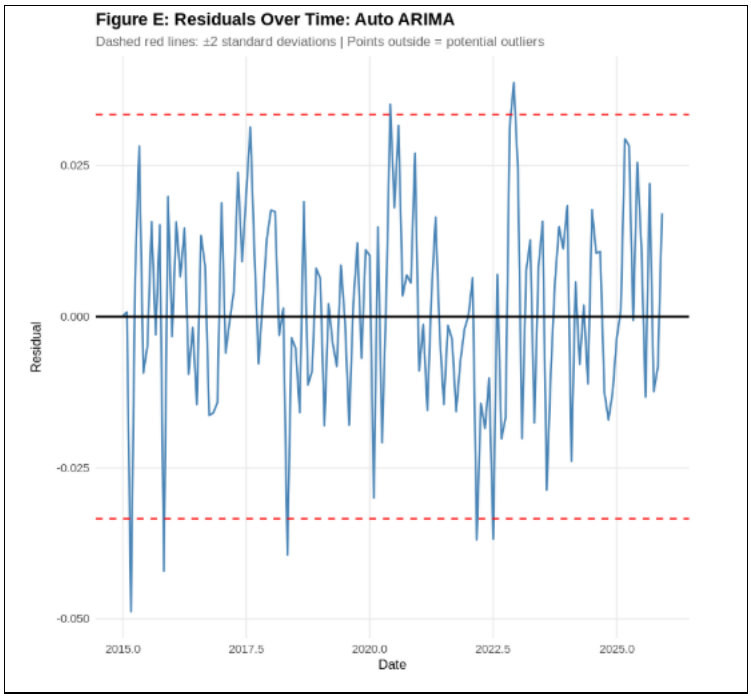

Figure 10 : Residuals over time for Auto-ARIMA

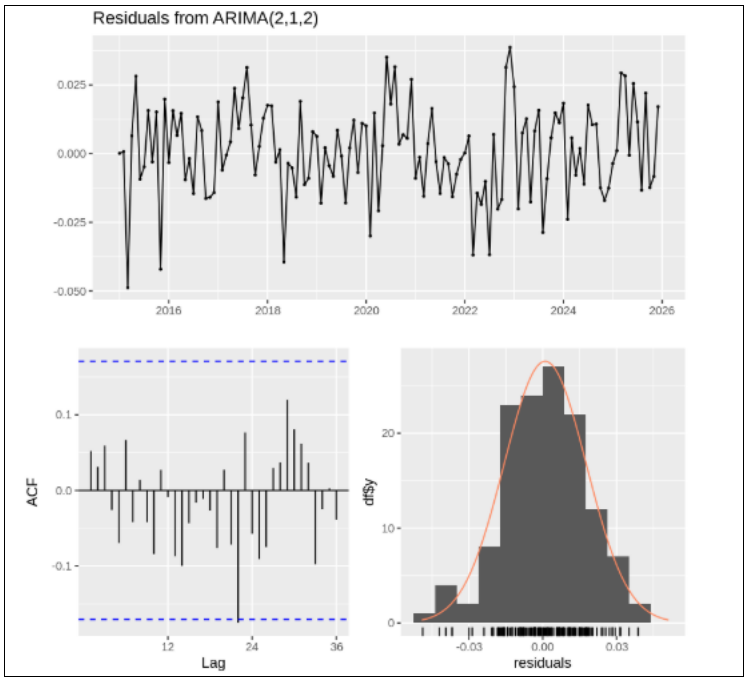

Figure 11:. Residual and lag for Auto-ARIMA

The RMSE as 0.016676 is the lowest of all eight models confirming the best in-sample fit. All eight models have passed the Ljung-Box white noise test at lag 12 as shown in **Figure 12,** which means that the more complex models offer meaningful improvements in fit.

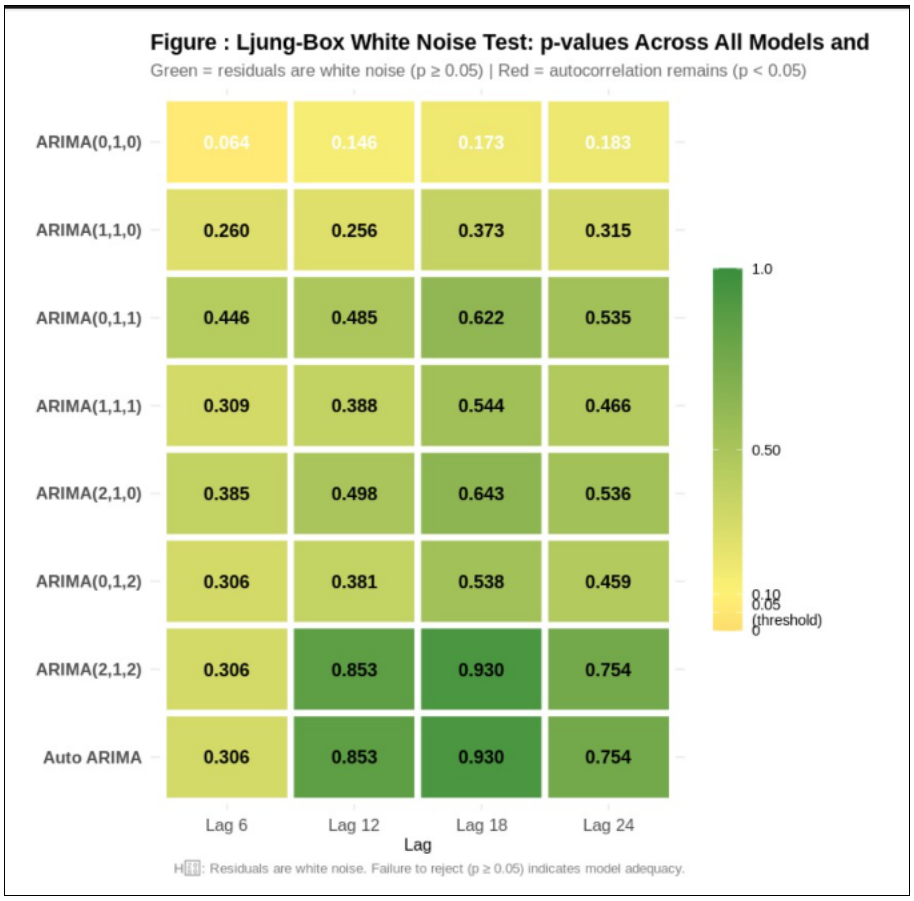

Figure 12:. Ljung-Box test to check white noise across all models

7.4 Forecasting with the Best Model

Using the verified ARIMA(2,1,2) model a 12-month ahead forecast was generated for the log real exchange rate over January to December 2026, with 80% and 95% prediction intervals. The forecast fan chart in **Figure 13** , and the monthly error bar chart in **Figure 14** . Table 11 presents the forecasts back transformed to the original RER scale via exponentiation.

The model predicts a stable exchange rate with slight fluctuations reflecting the AR and MA dependencies identified earlier. Forecast intervals are calculated using critical values from the normal distribution (e.g. approx. 1.96 for 95% confidence level). These intervals widen across the

next 12-month horizon from Jan 2026 to Dec 2026 that acknowledged the uncertainty in long-term predictions.

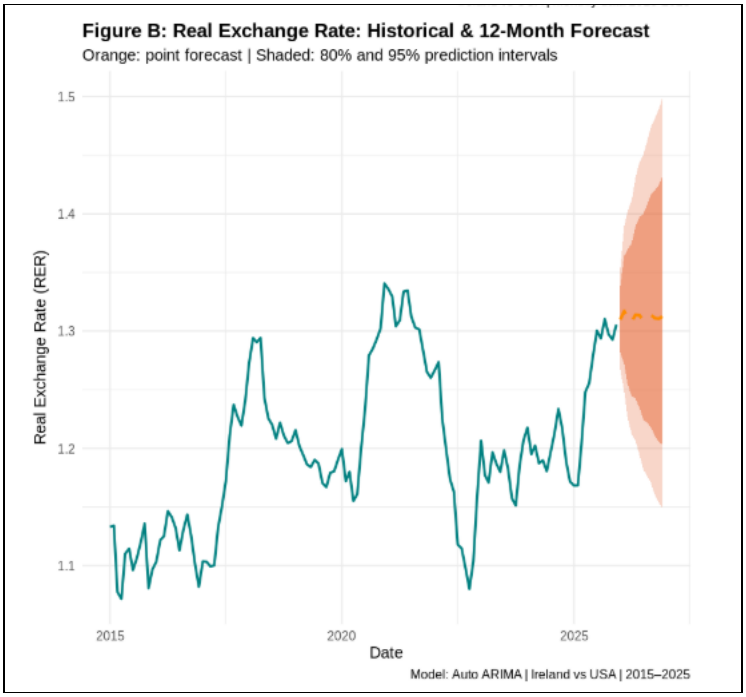

Figure 13 :. Real exchange rate: Historical and 12-month forecasts

The point forecasts are remarkably stable, ranging narrowly between 1.3081 and 1.3170 across all 12 months, suggesting that the ARIMA(2,1,2) model projects a broadly flat trajectory for the EUR/USD real exchange rate through 2026 at approximately 1.31. This is consistent with the model's mean-reverting dynamics identified in the AR coefficients. However, the 95% prediction intervals widen substantially as the horizon extends, from 0.044 in January 2026 to 0.176 (both positive and negative) by December 2026. This widening reflects the compounding of uncertainty over time, a fundamental property of ARIMA forecasting (Hamilton, 1994) and underscores that while the point estimate is stable the probabilistic range of outcomes is considerable at longer horizons.

To check the predictive performance the model was estimated again on a training set of 120 months (January 2015 to December 2024) and 12-month ahead forecasts were generated for the hold-out test period of January to December 2025. Refer Table 12 for accuracy results.

Out of Sample Forecast for Future 12-Month forecast, 2025

```
Training set: Jan 2015 – Dec 2024 (120 obs)
Test set:     Jan 2025 – Dec 2025 (12 obs)

                   ME     RMSE      MAE       MPE      MAPE     MASE     ACF1
Training set 0.000253 0.016579 0.013262 -0.459247  9.100618 0.236287 0.014104
Test set     0.070544 0.081500 0.072175 27.875300 28.923474 1.285953 0.742349
             Theil's U
Training set        NA
Test set      3.779082
```

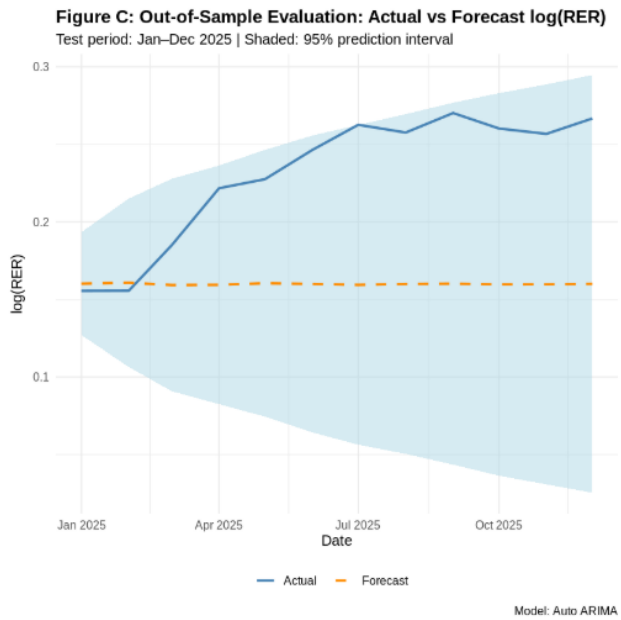

Figure 14 :. Out of sample evaluation: Actual vs Forecast – Log(RER)

In-sample performance is strong when RMSE is shown as 0.01658 and MAPE as 9.10% and this confirms a well-fitted model on training data. The in-sample MASE of 0.236 indicates the model is approximately 76% more accurate than the naïve random walk benchmark on the training data. The story for the out-of-sample results are different. The test set RMSE as 0.0815 and MAPE as 28.92% represents a substantial decline from in-sample performance. The MASE = 1.286 indicates the model performs worse than the naive random walk benchmark in the hold-out period. Theil's

U of 3.779 similarly confirms the out-of-sample forecast is considerably less accurate than a no-change prediction.

```
Overall Model Ranking (by average normalised score):
        Model AIC_score BIC_score RMSE_score LB_score Overall
   Auto ARIMA     1.000     0.000      1.000    1.000   0.750
 ARIMA(2,1,2)     1.000     0.000      1.000    1.000   0.750
 ARIMA(0,1,1)     0.922     1.000      0.598    0.480   0.750
 ARIMA(2,1,0)     0.776     0.468      0.636    0.497   0.594
 ARIMA(1,1,0)     0.779     0.835      0.525    0.155   0.573
 ARIMA(1,1,1)     0.704     0.384      0.599    0.342   0.507
 ARIMA(0,1,2)     0.705     0.385      0.599    0.332   0.505
 ARIMA(0,1,0)     0.000     0.295      0.000    0.000   0.074
```

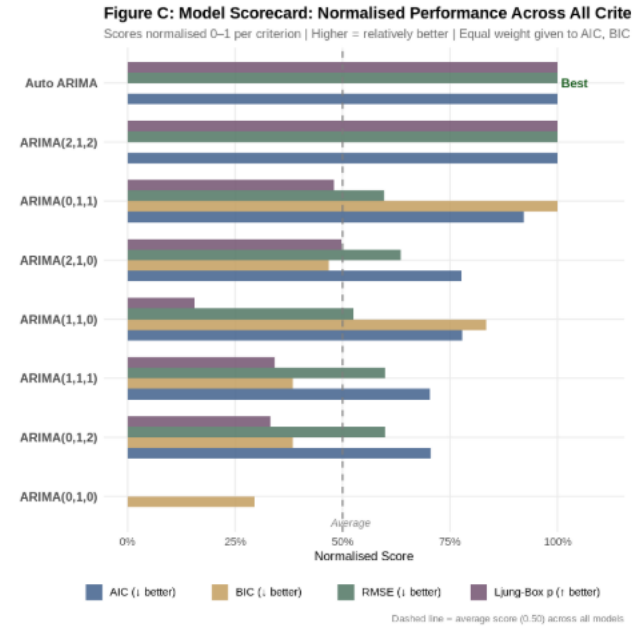

Figure 15 :. Overall model ranking and scorecard

```
12-Month Ahead Forecasts - log(RER):
         Point Forecast     Lo 80     Hi 80     Lo 95     Hi 95
Jan 2026      0.2698535 0.2480654 0.2916416 0.2365315 0.3031755
Feb 2026      0.2753797 0.2403611 0.3103984 0.2218234 0.3289361
Mar 2026      0.2705988 0.2263923 0.3148052 0.2029908 0.3382067
Apr 2026      0.2685677 0.2187444 0.3183910 0.1923695 0.3447658
May 2026      0.2730102 0.2172457 0.3287747 0.1877257 0.3582947
Jun 2026      0.2725875 0.2105767 0.3345982 0.1777502 0.3674248
Jul 2026      0.2694652 0.2027229 0.3362075 0.1673917 0.3715388
Aug 2026      0.2711394 0.2001454 0.3421333 0.1625634 0.3797153
Sep 2026      0.2727341 0.1970508 0.3484174 0.1569865 0.3884817
Oct 2026      0.2707938 0.1908651 0.3507224 0.1485534 0.3930341
Nov 2026      0.2704518 0.1868752 0.3540283 0.1426324 0.3982711
Dec 2026      0.2720447 0.1847026 0.3593869 0.1384665 0.4056230

Back-Transformed Forecasts: Original RER Scale
   Month LogRER_Pt RER_Point RER_Lo80 RER_Hi80 RER_Lo95 RER_Hi95
 2026-01    0.2699    1.3098   1.2815   1.3386   1.2668   1.3542
 2026-02    0.2754    1.3170   1.2717   1.3640   1.2484   1.3895
 2026-03    0.2706    1.3107   1.2541   1.3700   1.2251   1.4024
 2026-04    0.2686    1.3081   1.2445   1.3749   1.2121   1.4117
 2026-05    0.2730    1.3139   1.2426   1.3893   1.2065   1.4309
 2026-06    0.2726    1.3134   1.2344   1.3974   1.1945   1.4440
 2026-07    0.2695    1.3093   1.2247   1.3996   1.1822   1.4500
 2026-08    0.2711    1.3115   1.2216   1.4079   1.1765   1.4619
 2026-09    0.2727    1.3136   1.2178   1.4168   1.1700   1.4747
 2026-10    0.2708    1.3110   1.2103   1.4201   1.1602   1.4815
 2026-11    0.2705    1.3106   1.2055   1.4248   1.1533   1.4892
 2026-12    0.2720    1.3126   1.2029   1.4325   1.1485   1.5002
```

Figure 16 :. Future 12 month ahead forecasts

An ARIMA model has no process to incorporate future monetary policy signals, and it can only extrapolate the linear autocorrelation patterns of the past. In case the pattern breaks as they invariably do around monetary policy turning points, the model's out-of-sample accuracy collapses exactly as Meese and Rogoff *(1983: 22)* predicted four decades ago.

8. CONCLUSION

Our study examined the EUR/USD exchange rate dynamics between Ireland and the United States over the period January 2015 to December 2025, employing three analytical frameworks: time series properties analysis, Purchasing Power Parity testing, and Box-Jenkins forecasting.

The preliminary analysis confirmed that all four log-transformed variables are integrated of order one, I(1), satisfying the preconditions for cointegration testing and ensuring that subsequent regressions are not spurious *(Granger and Newbold, 1974: 111)*. Both absolute and relative PPP were decisively rejected at the monthly frequency. The failure of absolute PPP is consistent with the Balassa-Samuelson structural argument Ireland's price level is permanently distorted by the productivity of US multinational activity meaning the rejection is not a finite-sample limitation but a structural economic reality *(Balassa, 1964)*. The failure of relative PPP, evidenced by a slope coefficient of -0.150 with $R^2 = 0.002$, reflects the well-known dominance of financial account transactions in short-horizon EUR/USD determination. Monthly exchange rate movements are driven by interest rate differentials, speculative positioning, and central bank forward guidance, forces that overwhelm the goods-market arbitrage mechanism that PPP assumes. This finding is entirely consistent with *Meese and Rogoff's (1983: 3)* foundational result that exchange rate movements at short horizons are effectively orthogonal to macroeconomic fundamentals, and with *Rogoff's (1996: 647)* subsequent documentation that real exchange rate half-lives of three to five years imply PPP can only be expected to hold over multi-decade samples, far longer than the eleven years available here.

The Box-Jenkins modelling exercise identified **ARIMA(2,1,2)** as the optimal specification by AIC, with all four coefficients statistically significant and residuals confirmed as white noise across all diagnostic tests. The in-sample RMSE of 0.01658 and MASE of 0.236 confirm strong

training-period fit. However, the out-of-sample performance during 2025 MASE of 1.286, Theil's U of 3.779 demonstrates that the model performs worse than the naïve random walk benchmark when faced with genuinely new data. This result is not a failure of methodology. It is a confirmation of the Meese-Rogoff puzzle: no univariate time series model can systematically outperform the random walk for exchange rate forecasting out-of-sample, because the dominant drivers of exchange rates are forward-looking monetary policy expectations that are absent from historical autocorrelation patterns *(Meese and Rogoff, 1983: 22).* The 2025 test period was characterised by a sharp ECB-Fed monetary policy divergence that represented a structural regime shift from the training period, exposing the fundamental limitation of any ARIMA model estimated on one monetary regime and asked to forecast through another.

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AI Appendix:

- AI tools (Claude, ChatGPT and Gemin) were used for proofreading assistance only. All analysis, interpretation and conclusions are the work of the authors.